# CineTracker carrier board v0.62

Power

File: power.kicad_sch

Genlock

File: genlock.kicad_sch

Audio

File: audio.kicad_sch

Encoders

File: encoders.kicad_sch

UI

File: ui.kicad_sch

Each block is one sheet. Nets that cross sheets use global labels (firmware names: I2S_*, SPI_*, WIRE_*, ...).

Rails use power symbols: GND, +3V3 (from the Teensy), +5V (J7 / PoE board), +3V3A and +5VA (ferrite-filtered analog).

Source of truth for pin functions: firmware in this repository (include/*.h, src/aes67.cpp).

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Stack: carrier (bottom) / Teensy 4.1 (components up) / PoE board (top)

Teensy 4.1 carrier: genlock, LTC audio, 3 lens encoders, status UI

**CineTracker (github.com/dkontiscom/cinetracker)**

Sheet: /

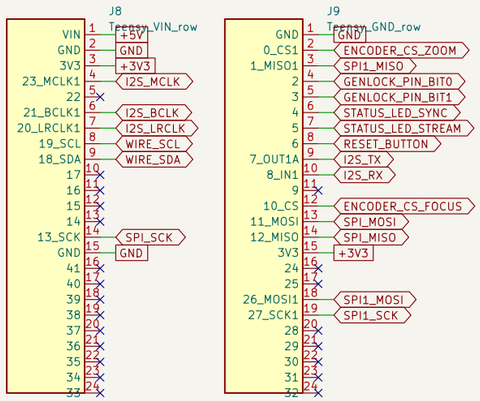
File: cinetracker-carrier.kicad_sch

***Title: CineTracker carrier board***

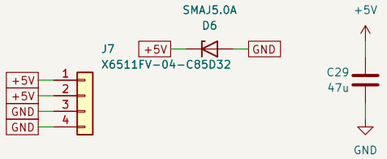
| Size: A4 | Date: 2026-10-13 | **Rev: v0.62** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 1/6 |

# Power and Teensy 4.1

## Teensy 4.1 sockets (USB end = pin 1; VIN row on top, GND/0/1 row below)



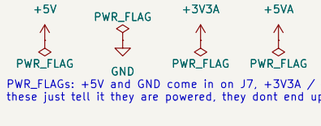

## 5 V input from the PoE board (J7) and protection





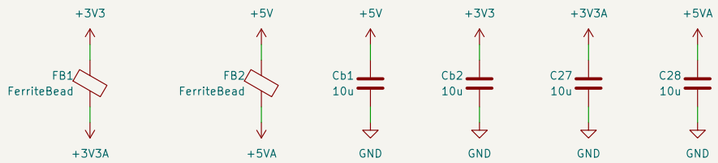





PWR_FLAGs: +5V and GND come in on J7, +3V3A / +5VA through FB1 / FB2, so ERC cant see a driver.
these just tell it they are powered, they dont end up on the board

## Analog rails and bulk decoupling




## Mounting holes




MH5/MH6 (v0.5e): M2.5 standoff holes for the PoE board v0.1, NPTH, no copper within 3.2 mm. Use nylon standoffs.

Teensy 4.1 rows CORRECTED in v0.5d. v0.3 to v0.5c had the two rows swapped and the VIN-row order wrong
(it put +5 V on the Teensy GND pin). Source: PJRC pinout card rev4 and dimension drawing, top view, USB on the left:
top row VIN GND 3.3V 23..13 GND 41..33, bottom row GND 0..12 3.3V 24..32.
J7 receives 5 V from the PoE board (or a bench supply). CUT the Teensy VUSB-VIN pads (PJRC) so USB carries data only.
D6 SMAJ5.0A clamps transients only: its 6.4-7.0 V breakdown is above the Teensy LDO 6.0 V abs max.

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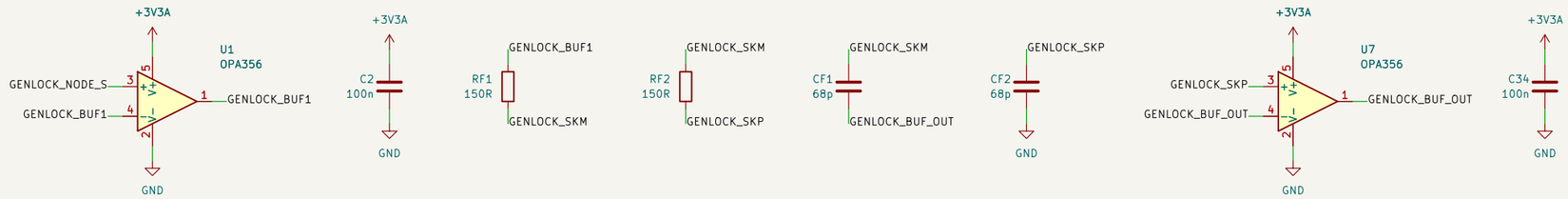
| | | |
|---|---|---|
| Sheet: /Power/ | | |
| File: power.kicad_sch | | |
| ***Title: CineTracker carrier: Power and Teensy 4.1*** | | |
| Size: A3 | Date: 2026-10-13 | **Rev: v0.62** |
| KiCad E.D.A. 10.0.7 | | Id: 2/6 |

# Genlock: tri-level sync output

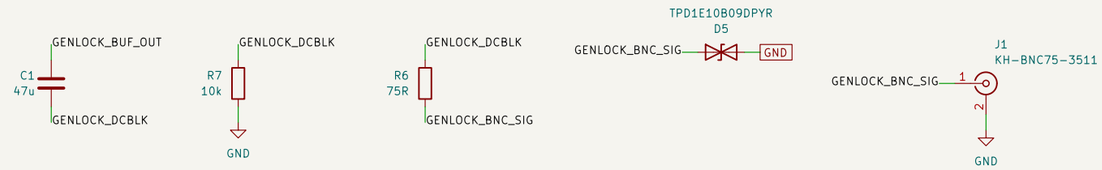
## 2-bit R-2R ladder (BIT1 = MSB, pin 3; BIT0 = LSB, pin 2) with offset




## Buffer, 2-pole Sallen-Key low-pass (Q = 0.5), line driver




## DC block, 75 ohm source, ESD, BNC




Firmware t41-tri-sync.h drives (BIT1,BIT0): 01 = blank 0 mV, 00 = -300 mV, 10 = +300 mV at the camera (75 ohm cable into 75 ohm).
R15 5k6 to +3V3A || R5 5k1 to GND lifts the buffer levels to 0.43 / 1.03 / 1.63 V (inside OPA356 CM range, V+ - 1.5 V).
Sallen-Key RF 150 ohm / CF 68 pF 1% C0G, Q = 0.5: 10-90% rise 35.2 / 35.3 / 35.2 ns in SPICE (TI OPA356 model),
inside both ST 274 windows (24/25/30p 33.7-74.1 ns, 50/60p 16.8-37.0 ns). See documentation/genlock-simulation.md.
Trailing edge needs BIT1-low and BIT0-high in one store: firmware writes them with one GPIO9 DR_TOGGLE store (commit ab11050).

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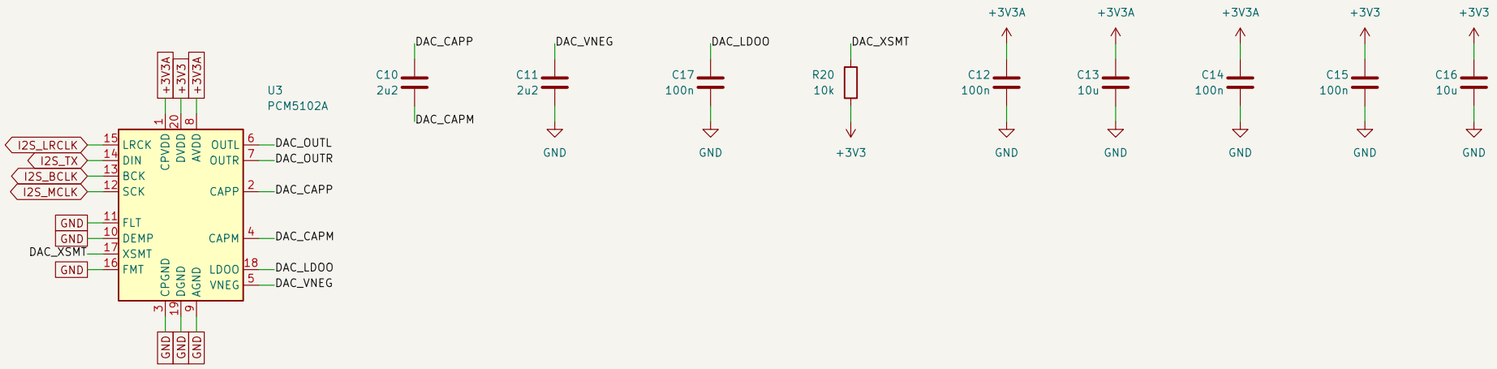
Sheet: /Genlock/
File: genlock.kicad_sch

***Title: CineTracker carrier: Genlock: tri-level sync output***

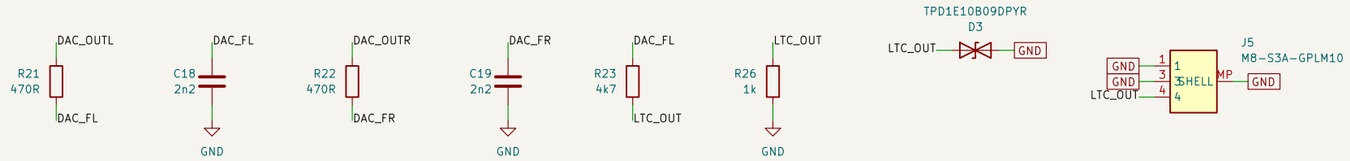
| | | |
|---|---|---|
| Size: A3 | Date: 2026-10-13 | **Rev: v0.62** |
| KiCad E.D.A. 10.0.7 | | Id: 3/6 |

# Audio: LTC timecode in/out

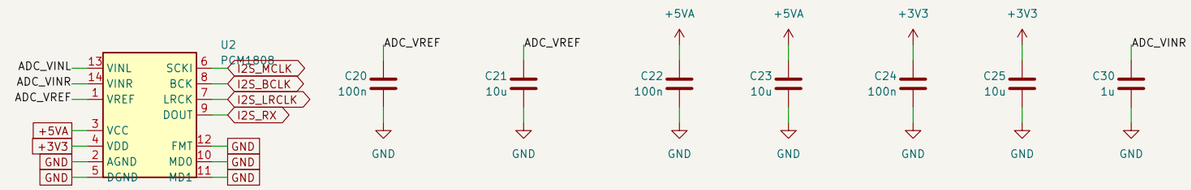
## DAC: PCM5102A (I2S slave, no control interface)



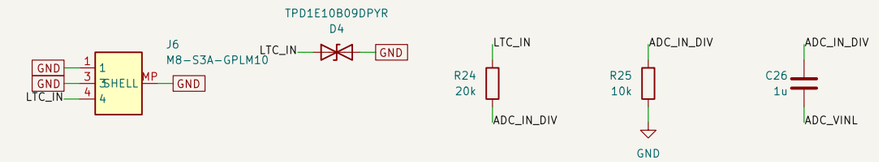

## DAC output filter and LTC out level




## ADC: PCM1808 (slave, I2S, MD0 = MD1 = FMT = 0)




## LTC in level and coupling




Mono LTC on the left channels only (firmware aes67.cpp, channel 0). OUTR / VINR terminated.

LTC out: about 0.96 Vpp open circuit, about 0.84 kohm source. LTC in: about 10 Vpp to ADC full scale, about 28.6 kohm.

Level choices are design choices (no public LTC level spec checked). Change R23/R26 and R24/R25 to suit your gear.

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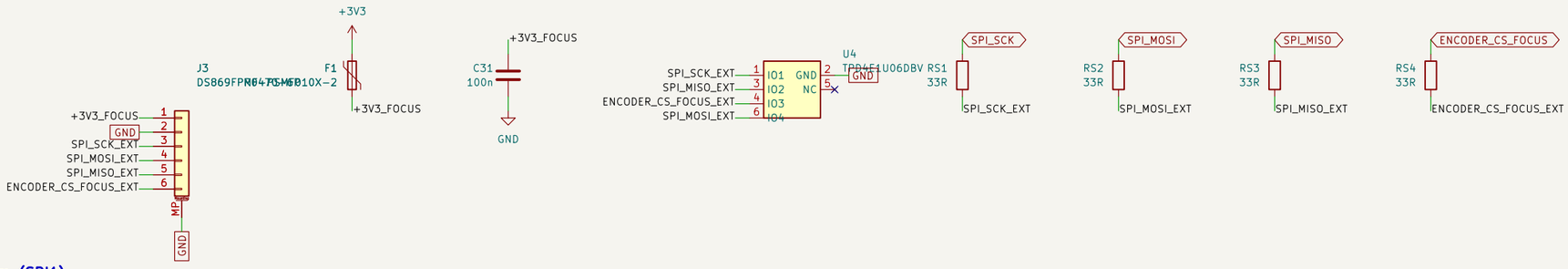
Sheet: /Audio/
File: audio.kicad_sch

***Title: CineTracker carrier: Audio: LTC timecode in/out***

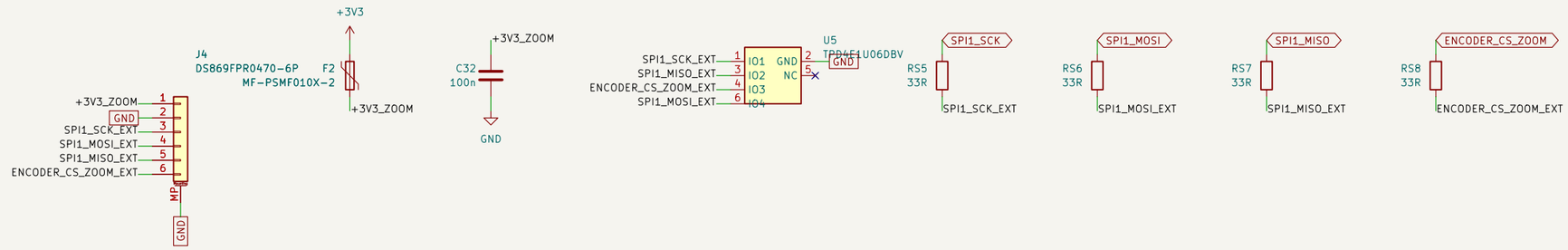
| Size: A3 | Date: 2026-10-13 | **Rev: v0.62** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 4/6 |

# Lens encoders: focus, zoom (SPI), iris (I2C)

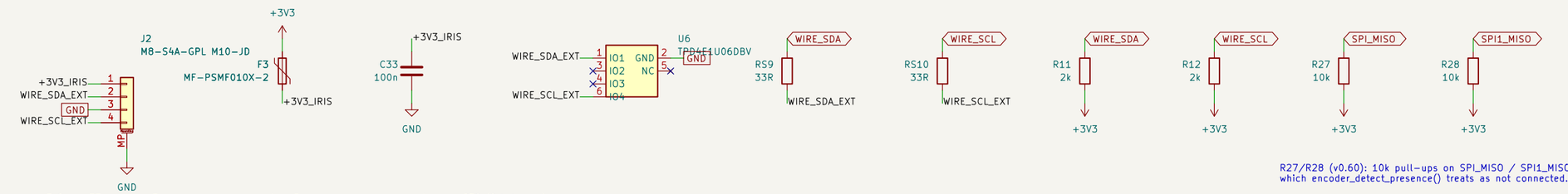
## Focus (SPI)




## Zoom (SPI1)




## Iris (I2C, AS5048B)




Connector -> TVS -> 33 ohm -> Teensy on every cable line. Each encoder supply has its own PTC (0.1 A hold).

AS5048A over SPI at 8 MHz MODE1 (focus on SPI, zoom on SPI1); AS5048B over I2C at 400 kHz (iris), 2 kohm pull-ups.

M8 sockets are female: cordsets need a male M8 plug at the board end.

R27/R28 (v0.60): 10k pull-ups on SPI_MISO / SPI1_MISO. A missing encoder reads 0xFFFF (error flag set), which encoder_detect_presence() treats as not connected.

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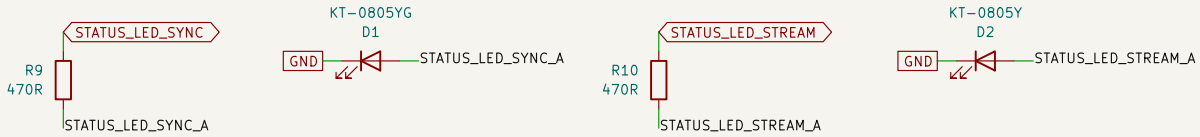
Sheet: /Encoders/
File: encoders.kicad_sch

***Title: CineTracker carrier: Lens encoders: focus, zoom (SPI), iris (I2C)***

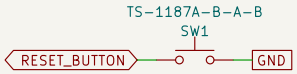
| | | |
|---|---|---|
| Size: A3 | Date: 2026-10-13 | **Rev: v0.62** |
| KiCad E.D.A. 10.0.7 | | Id: 5/6 |

# Status LEDs and reset button

## Status LEDs (active high)




## Reset / factory-reset button (INPUT_PULLUP in firmware)




LED pins 4 and 5 (active high). Reset pin 6, INPUT_PULLUP, button to GND; hold 5 s = factory reset.

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Sheet: /UI/
File: ui.kicad_sch

***Title: CineTracker carrier: Status LEDs and reset button***

| Size: A4 | Date: 2026-10-13 | **Rev: v0.62** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 6/6 |